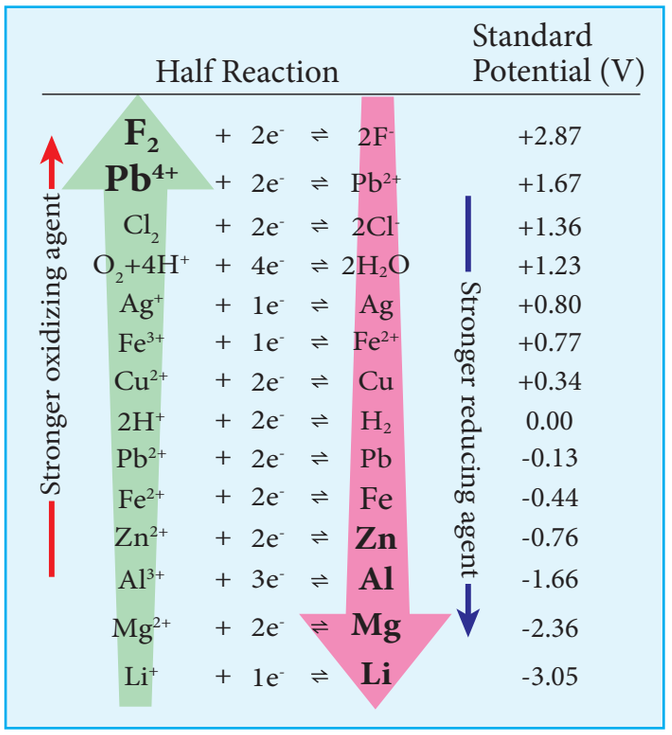
Electrochemical series

We have already learnt that the standard single electrode potentials are measured using standard hydrogen electrode. The standard electrode potential at 298K for various metal - metal ion electrodes are arranged in the decreasing order of their standard reduction potential values as shown in the figure.



This series is called electrochemical series.

The standard reduction potential (E°) is a measure of the oxidising tendency of the species. The greater the E° value, greater is the tendency shown by the species to accept electrons and undergo reduction. So higher the (E°) Value, lesser is the tendency to undergo corrosion

EVALUATION

Choose the correct answer:

1. The number of electrons that have a total charge of 9650 coulombs is

   a) 6.22×10^{23} b) 6.022×10^{24}

   c) 6.022×10^{22} c) 6.022×10^{-34}

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2. Consider the following half cell reactions:

   $Mn^{2+} + 2e^- \rightarrow Mn \quad E^\circ = -1.18V$

   $Mn^{2+} \rightarrow Mn^{3+} + e^- \quad E^\circ = -1.51V$

   The E° for the reaction $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$, and the possibility of the forward reaction are respectively.

   a) 2.69V and spontaneous b) -2.69 and non spontaneous

   c) 0.33V and Spontaneous d) 4.18V and non spontaneous

3. The button cell used in watches function as follows

   $Zn\,(s) + Ag_2O\,(s) + H_2O\,(l) \rightleftharpoons 2\,Ag\,(s) + Zn^{2+}(aq) + 2OH^-\,(aq)$ the half cell potentials are $Ag_2O\,(s) + H_2O\,(l) + 2e^- \rightarrow 2Ag\,(s) + 2\,OH^-\,(aq) \quad E^\circ = 0.34V$ and $Zn\,(s) \rightarrow Zn^{2+}(aq) + 2e^- \quad E^0 = 0.76\,V$. The cell potential will be

   a) 0.84V b) 1.34V c) 1.10V d) 0.42V

XII U9 Electro Chemistry.indd 62 11-01-2021 14:44:36